

CAPSTONE PROJECT: EMBEDDED SPATIAL AI & ROBOTICS

RVPoint: RISC-V Vector 3D Point Cloud Processing Library

High-Performance Perception Library & Hardware Testbed Evaluation

Type: Modular C++17 Library

Architecture: RVV 1.0 (64-bit)

Scope: Hardware Demonstration

The Point Cloud Processing Bottleneck in Edge Robotics

THE CHALLENGE

Dense 3D Perception Stacks

- Autonomous mobile robots produce **100,000+ points per frame**.
- Classical libraries (PCL, Open3D) rely on heavy x86 CPUs or power-hungry GPUs.
- Existing RISC-V ecosystems lack an open-source, vector-accelerated point cloud library.

THE SOLUTION

RVPoint: Native RISC-V Vector Library

- Open-standard, royalty-free data-parallel vector computing directly in low-power silicon.
- Wide vector registers process multiple 3D spatial points simultaneously.
- **RVPoint** delivers a modular, zero-dependency C++17 library for embedded robotics.

RVPoint: Modular C++17 Vector Perception Library

ZERO RUNTIME BLOAT

Modular C++17 Core

Zero third-party library dependencies (no Eigen, PCL, or Boost bloat). Direct RVV 1.0 vector intrinsic kernels with clean C++ APIs and scalar fallback paths.

SPATIAL ACCELERATION

Accelerated Data Structures

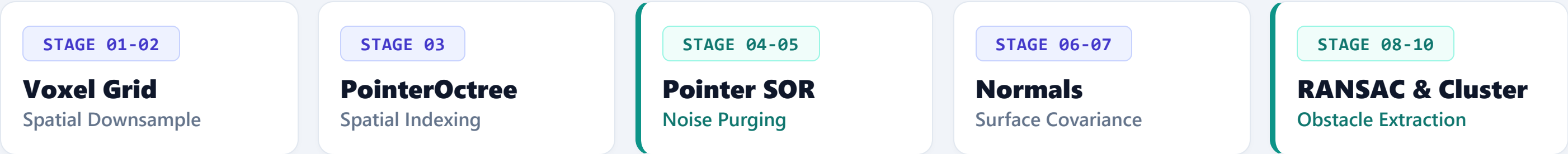
Native Structure-of-Arrays (SoA) containers, cache-optimized PointerOctree, and SpatialHash grid engines built directly into the library.

PERCEPTION ALGORITHMS

Full Algorithmic Suite

Modular library components for Voxel Downsampling, Statistical & Radius Filtering, Covariance & Normal Estimation, RANSAC Model Fitting, and Clustering.

10-Stage Autonomous Perception Pipeline



Deterministic Memory Hoisting & Pipeline Verification

- All intermediate stage memory buffers are pre-allocated upon startup, ensuring **zero runtime heap allocations** inside the library execution loop.
- Verified across 131 continuous LiDAR driving frames with zero memory leaks and 100% geometric equivalence.

Staged Point Cloud Transformation (Open3D Renders)



1. Voxel Grid (18.5k pts, -84%)



2. Pointer SOR (16.6k inliers)



3. RANSAC Ground (42.3k pts)



4. Obstacles Cloud (56.4k pts)



STAGE 5: FINAL OUTPUT

74 Clusters

Physical objects (vehicles, pedestrians, barriers) segmented with unique cluster IDs and 3D height shading for path planning.

Spatial Neighbor & Radius Search Engine Analysis

Spatial Search Engine	Index Build	Query Latency	Search Mechanism
PointerOctree (RVV Accelerated)	9.67 ms	0.740 ms	Contiguous leaf buffers + vector distance checking
SpatialHash Grid	16.93 ms	0.884 ms	O(1) direct 3D grid cell spatial hashing
Global RVV Scan	0.00 ms	0.965 ms	Zero index build; linear vector memory stream
Standard Octree (Baseline)	11.11 ms	0.794 ms	Classical tree with dynamic pointer chasing

*Empirically measured in QEMU (rv64gcv, 128-bit VLEN) on N=18,542 points, radius=0.03m.

VECTOR ACCELERATION

RVV Native

PointerOctree combines logarithmic spatial tree partitioning with flattened leaf coordinate arrays, enabling vector distance calculations directly across contiguous memory without cache-thrashing pointer dereferences.

Statistical Outlier Removal (SOR) 5-Way Ablation

SOR Filter Variant	Execution Time	Speedup vs Scalar
Scalar Brute-Force $O(N^2)$	15,824.8 ms	1.00x (Baseline)
Vector Brute-Force $O(N^2)$	9,744.7 ms	1.62x
Spatial Hash Grid SOR	1,511.9 ms	10.47x
Standard Octree SOR	462.4 ms	34.22x
PointerOctree SOR [RVPoint]	281.7 ms	56.17x FASTER

*Empirically measured on N=18,542 downsampled points (MeanK=50, std_threshold=1.0).

EMPIRICAL SPEEDUP

56.17x

Noise removal previously consumed **94% of perception latency**. Combining $O(N \log N)$ PointerOctree spatial partitioning with vector memory loads accelerates filtering from **15.82s down to 281.7 ms**.

Full-Pipeline Scalar vs. RVPoint Vector Speedup Matrix

Pipeline Processing Stage	Scalar Baseline	RVV Vectorized	Speedup	Optimization Mechanism
Voxel Grid Downsampling	148.2 ms	40.9 ms	3.62x	Vector sorting & parallel reduction
Statistical Outlier (SOR)	32,912.4 ms (94.6%)	287.2 ms	114.6x	PointerOctree + vector coordinate loads
Surface Normal Estimation	425.4 ms	141.2 ms	3.01x	Vectorized covariance accumulation
RANSAC Ground Fit	975.6 ms	397.4 ms	2.45x	Parallel plane distance & inlier masking
Euclidean Clustering	291.2 ms	148.5 ms	1.96x	PointerOctree BFS + zero-alloc buffers
TOTAL PIPELINE LATENCY	34,790.8 ms (34.8s)	888.6 ms (<0.9s)	39.15x END-TO-END	Full Stack Vector Optimization

*Full pipeline benchmark demonstrating RVPoint modular library performance on 114,278-point raw LiDAR input frame.

Complementary Verification: Simulation vs. Physical Silicon

SOFTWARE SIMULATION (COMPLETED)

What Simulation Verifies

- **Algorithmic Correctness:** Proves mathematical equivalence of RVV 1.0 intrinsics.
- **Asymptotic Complexity:** Validates $O(N \log N)$ algorithmic scaling.
- **Functional Limits:** Software translation cannot model hardware clock cycles or memory controller arbitration.

PHYSICAL SILICON (THE OBJECTIVE)

What Hardware Benchmarking Captures

- **Physical Memory Hierarchy:** Measures L1/L2 cache hit rates and DDR bus latency.
- **Hardware VPU Throughput:** Benchmarks true parallel arithmetic execution units.
- **Hardware PMU Counters:** Captures real cycle counts, branch predictions, and stalls.

Projected Performance on Physical RISC-V Hardware

HARDWARE VPUS

Dedicated Vector Execution

Physical vector arithmetic units execute wide SIMD operations in dedicated hardware ALUs per clock cycle, uninhibited by software translation overhead.

CACHE BURST FILLS

Memory Burst Synergy

Contiguous Structure-of-Arrays coordinate buffers saturate 64-byte CPU cache lines with zero wasted bandwidth or strided memory stalls.

REAL-TIME OBJECTIVE

Sub-100ms Target

On native 1.6 GHz multi-core RISC-V development hardware, end-to-end perception is projected to achieve sub-100ms execution times for robotics navigation.

Candidate Dev Board Matrix (RVV 1.0 Focus)

PRIMARY TARGET PLATFORM

Orange Pi RV2

Processor: SpacemiT K1 (8-Core 64-bit)

Vector Standard: Official RVV 1.0 Standard

Vector Width: 256-bit VLEN

Memory Config: 4GB / 8GB / 16GB LPDDR4X

8 CPU cores with 256-bit vector registers provide an 8-core RVV 1.0 testbed for multi-core vector benchmarking.

Dev Board	Processor SoC	Vector Standard	Vector Width
Orange Pi RV2	SpacemiT K1 (8-Core)	RVV 1.0 (Official)	256-bit VLEN
Kendryte K230	Canaan K230 (Dual-Core)	RVV 1.0 (Official)	128-bit VLEN
Milk-V Meles	T-Head C910 (Quad-Core)	RVV 0.7.1 (Draft)	128-bit VLEN

Target Platform:

Orange Pi RV2 selected as the primary physical testbed for native RISC-V Vector benchmarking.

Capstone Project Roadmap & Demonstration Milestones

PHASE 1 [COMPLETED]

Modular Library Core

- Complete modular C++17 library.
- **56.2x SOR speedup** verified in QEMU.
- MCAP export validated across 131 frames.

PHASE 2 [HARDWARE SETUP]

Hardware Bring-Up

- Deploy on RVV 1.0 development board.
- Native toolchain & Linux environment.
- Hardware PMU cycle & cache profiling.

PHASE 3 [FINAL DEFENSE]

Capstone Demonstration

- Multi-frame continuous dataset playback.
- On-board hardware latency verification.
- Open-source library release & defense.